

Quiz (Carolina Reaper)

Q1. What is the vertex of the graph of $y = |x - 2|$?

- (A) (0, 2)
- (B) (2, 0)
- (C) (2, 2)
- (D) (-2, 0)
- (E) (0, 0)

$-|x + 2|$?

- (A) Upward
- (B) Downward
- (C) Leftward
- (D) Rightward
- (E) None of the above

Q2. Which of the following is the graph of $y = |x + 1| - 3$?

- (A) A downward-facing V with vertex at (-1, -3)
- (B) An upward-facing V with vertex at (-1, -3)
- (C) A downward-facing V with vertex at (1, 3)
- (D) An upward-facing V with vertex at (1, -3)
- (E) A horizontal line at $y = -3$

Q6. If the graph of $y = |x|$ is reflected across the y-axis, what is the equation of the new graph?

- (A) $y = |x|$
- (B) $y = -|x|$
- (C) $y = |x + 1|$
- (D) $y = |x - 1|$
- (E) $y = -x$

Q3. What is the axis of symmetry of the graph of $y = |2x - 4|$?

- (A) $x = -2$
- (B) $x = 0$
- (C) $x = 1$
- (D) $x = 2$
- (E) $x = 4$

Q7. What is the vertex of the graph of $y = |x| + 2$?

- (A) (0, 0)
- (B) (0, 2)
- (C) (2, 0)
- (D) (-2, 0)
- (E) (0, -2)

Q4. If the graph of $y = |x|$ is shifted 3 units to the right, what is the equation of the new graph?

- (A) $y = |x - 3|$
- (B) $y = |x + 3|$
- (C) $y = x - 3$
- (D) $y = x + 3$
- (E) $y = -x + 3$

Q8. If the graph of $y = |x - 2|$ is shifted 2 units upward, what is the equation of the new graph?

- (A) $y = |x - 2| - 2$
- (B) $y = |x - 2| + 2$
- (C) $y = |x + 2| - 2$
- (D) $y = |x + 2| + 2$
- (E) $y = x - 2 + 2$

Q5. What is the direction of opening of the graph of $y =$

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 1| - 2$ and identify the vertex, axis of symmetry, and direction of opening.

Q10. Suppose the graph of $y = |x|$ is transformed into the graph of $y = |x + 2| - 3$. Describe the transformations that occurred and graph the new function.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the properties of absolute value functions.
- Tip 2: Encourage students to use graphing calculators or software to visualize the transformations.
- Tip 3: Remind students to check their work and verify their answers.